

EE227 – Digital Logic Design

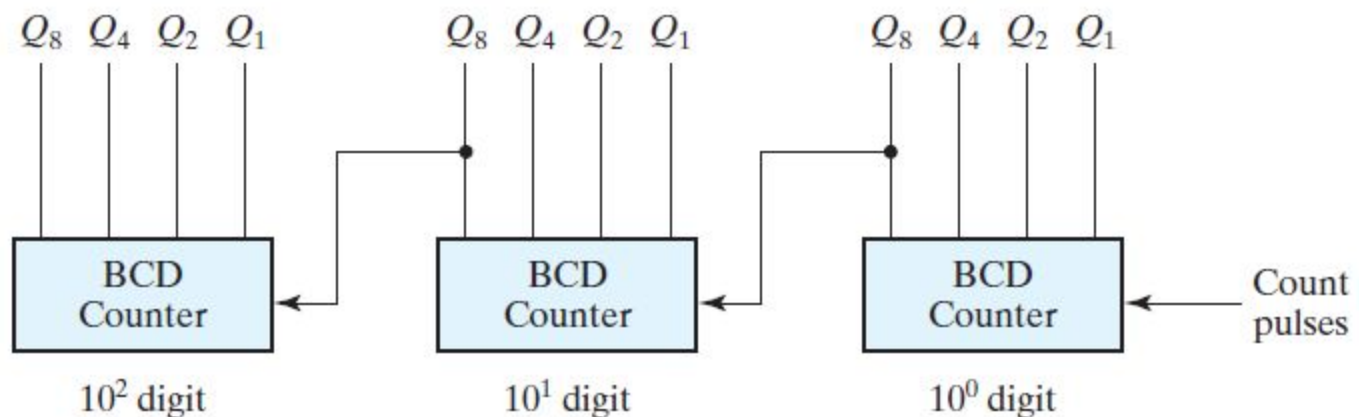
Lecture 27

Outline

- BCD Decade Counter
- Synchronous Counter
 - BCD Counter
 - 4 bit Binary Counter with Parallel Load
- Quiz 7

BCD Decode Counter

- The BCD counter is a decade counter as it counts from 0 to 9
- To count in decimal from 0 to 99, we need a two-decade counter
- To count from 0 to 999, we need a three-decade counter
- Multiple decade counters can be constructed by connecting BCD counters in cascade, one for each decade



Synchronous Counters

- Synchronous counters are different from ripple counters in that clock pulses are applied to the inputs of all flip-flops
- A common clock triggers all flip-flops simultaneously, rather than one at a time in succession as in a ripple counter

Synchronous BCD Counter (1/4)

- A BCD counter counts in binary-coded decimal from 0000 to 1001 and back to 0000
- Because of the return to 0 after a count of 9, a BCD counter does not have a regular pattern, unlike a straight binary count
- In a BCD counter we have an output y , which is equal to 1 when the present state is 1001
- In this way, y can enable the count of the next-higher significant decade while the same pulse switches the present decade from 1001 to 0000

Synchronous BCD Counter (2/4)

Present State				Next State				Output	Flip Flop Inputs			
Q ₈	Q ₄	Q ₂	Q ₁	Q ₈	Q ₄	Q ₂	Q ₁	y	TQ ₈	TQ ₄	TQ ₂	TQ ₁
0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	1	0	0	1	0	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	0	1
0	0	1	1	0	1	0	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	0	1
0	1	0	1	0	1	1	0	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	0	1
0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	0	1
1	0	0	1	0	0	0	0	1	1	0	0	1
1	0	1	0	X	X	X	X	X	X	X	X	X
1	0	1	1	X	X	X	X	X	X	X	X	X
1	1	0	0	X	X	X	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X	X	X	X

Synchronous BCD Counter (3/4)

		Q_2Q_1			
		00	01	11	10
Q_8Q_4	00				
	01				
	11	X	X	X	X
	10		1	X	X

$y = Q_8Q_1$

		Q_2Q_1			
		00	01	11	10
Q_8Q_4	00				
	01			1	
	11	X	X	X	X
	10		1	X	X

$TQ_8 = Q_8Q_1 + Q_4Q_2Q_1$

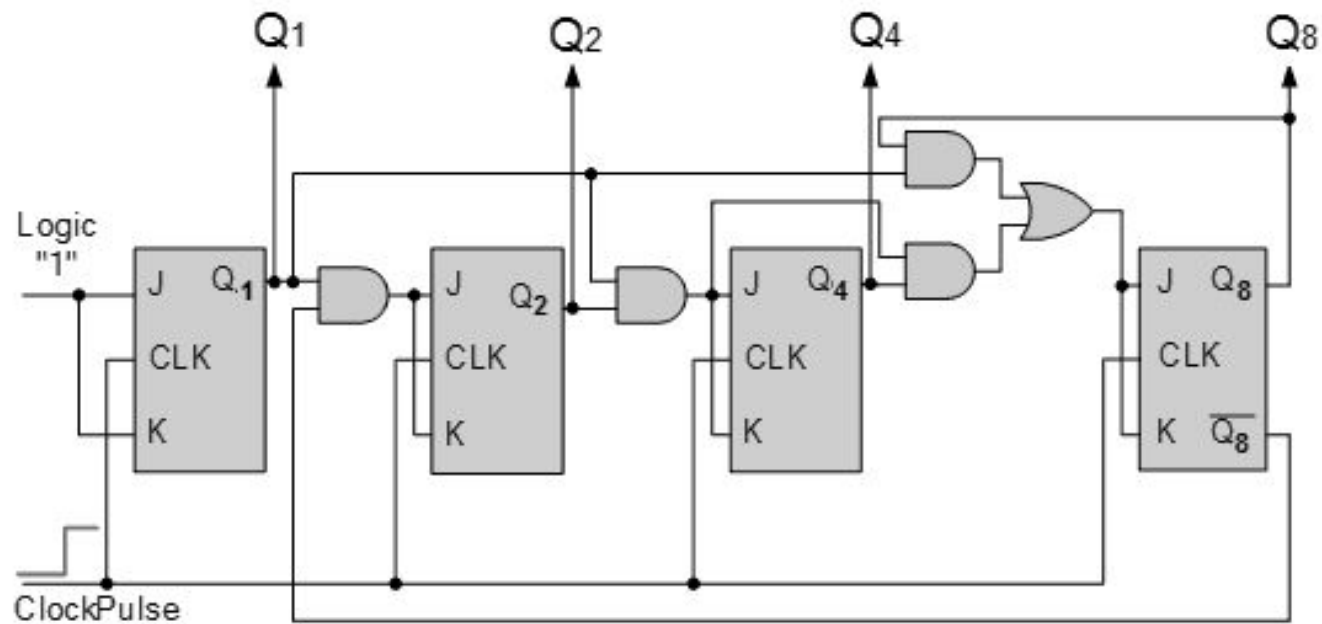
		Q_2Q_1			
		00	01	11	10
Q_8Q_4	00			1	
	01			1	
	11	X	X	X	X
	10			X	X

$TQ_4 = Q_2Q_1$

		Q_2Q_1			
		00	01	11	10
Q_8Q_4	00		1	1	
	01		1	1	
	11	X	X	X	X
	10			X	X

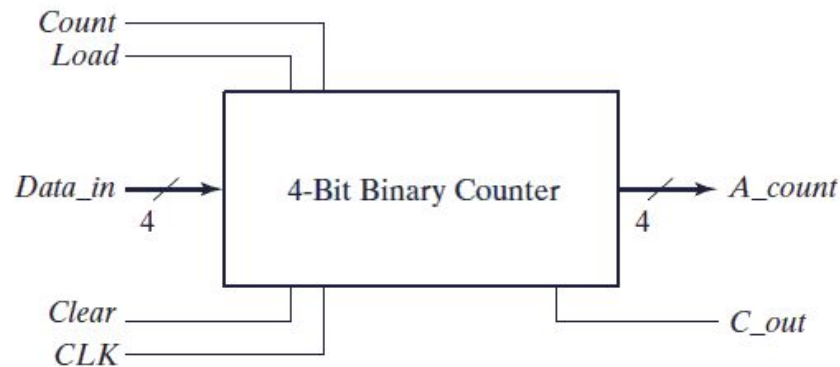
$TQ_2 = Q'_8Q_1$

Synchronous BCD Counter (4/4)



Binary Counter with Parallel Load

- Counters employed in digital systems quite often require a parallel-load capability for transferring an initial binary number into the counter prior to the count operation
- The block diagram of a 4 bit binary counter with parallel load is shown below

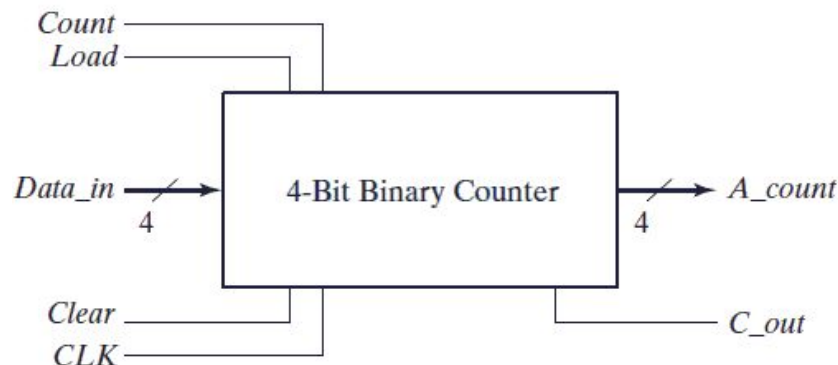


- The carry output becomes a 1 if all the flip-flops are equal to 1 while the count input is enabled
- The carry output is useful for expanding the counter to more than four bits

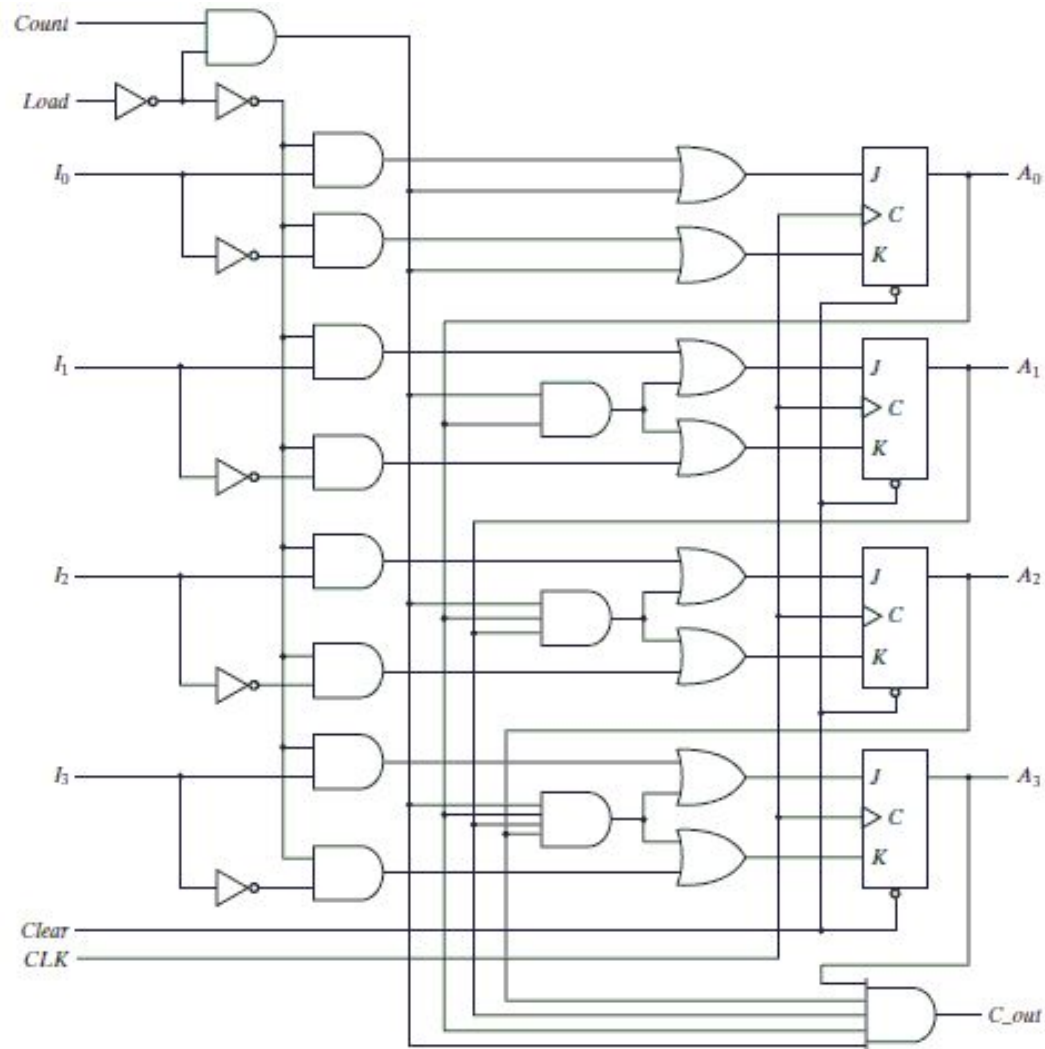
Binary Counter with Parallel Load

- The operations of a 4 – bit binary counter with parallel load is summarized in the table below

Clear	CLK	Load	Count	Function
0	X	X	X	Clear to 0
1	↑	1	X	Load inputs
1	↑	0	1	Count next binary state
1	↑	0	0	No change

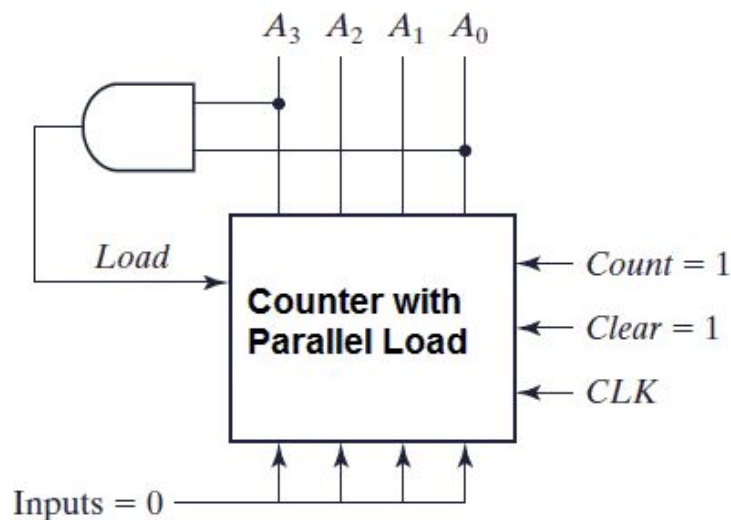


Binary Counter with Parallel Load

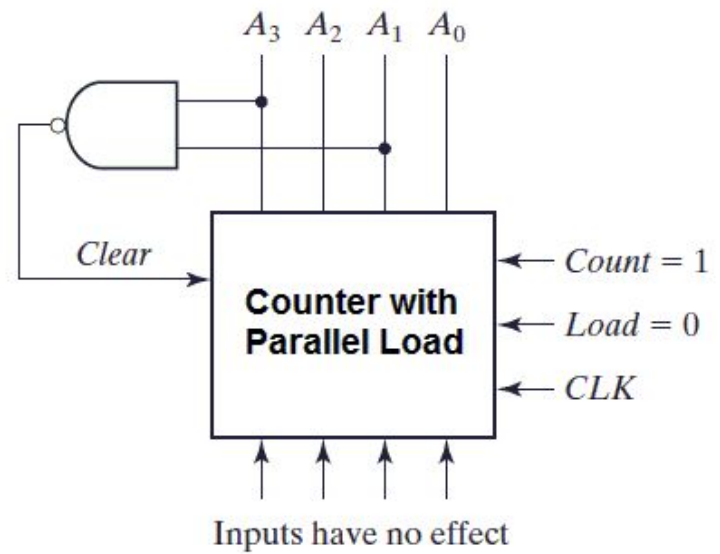


Uses of Binary Counter with Parallel Load

- A counter with a parallel load can be used to generate any desired count sequence
- Using as a BCD counter



(a) Using the load input



(b) Using the clear input